Practical - 12

steps

1. **Open VS Code:** Create or open a folder and create a Python file named tutorial12.py. 2. **Install Selenium:** Open the VS Code Terminal and run: pip install selenium 3. **Check Chrome:** Make sure Google Chrome is installed. Selenium Manager normally manages the required driver automatically in recent Selenium versions.
4. **Enter the code:** Copy the complete program shown in Section 4 into tutorial12.py and save the file.
5. **Run the program:** Click the Run button in VS Code, or run: python tutorial12.py

6. Python program

```
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.support.ui import WebDriverWait
from selenium.webdriver.support import expected_conditions as EC
import time

driver = webdriver.Chrome()
driver.maximize_window()
wait = WebDriverWait(driver, 60)

try:
    # Step 1: Open the website
    driver.get("https://www.saucedemo.com/")

    # Step 2: Enter username
    username = wait.until(
        EC.visibility_of_element_located((By.ID, "user-name"))
    )
    username.send_keys("standard_user")

    # Step 3: Enter password
    password = driver.find_element(By.ID, "password")
    password.send_keys("secret_sauce")

    # Step 4: Click Login
    login_button = driver.find_element(By.ID, "login-button")
    login_button.click()

    # Step 5: Wait for inventory page
    wait.until(EC.url_contains("inventory.html"))

    # Step 6: Validate login
    assert "inventory.html" in driver.current_url

    print("AI-Assisted Login Test: PASS")
    print("Requirement Validation: PASS")

    # Keep the browser open for 5 seconds
    time.sleep(5)

except Exception as e:
    print("AI-Assisted Login Test: FAIL")
    print("Error:", e)
finally:
    # Step 7: Close the browser
```

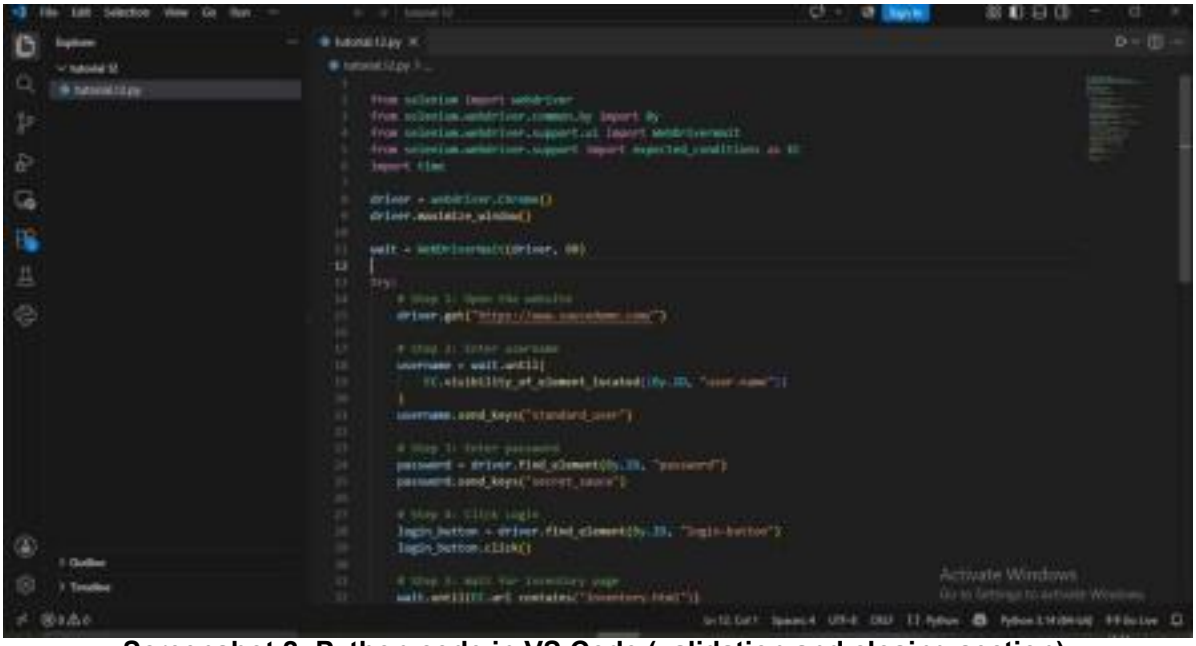
```
driver.quit()
```

Explanation of the code

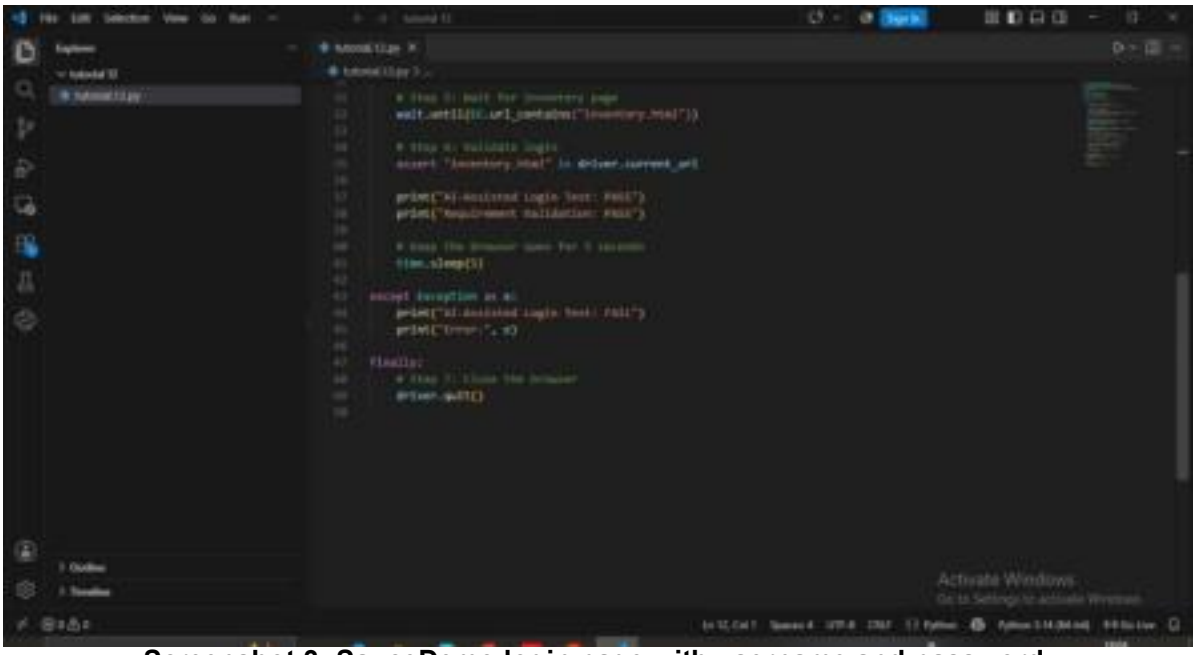
- **Import statements —** Load Selenium tools, element locators, explicit waits, expected conditions, and the time module.
- **webdriver.Chrome() —** Starts a Chrome browser controlled by Selenium.
- **WebDriverWait(driver, 60) —** Waits up to 60 seconds for a required condition instead of relying only on fixed delays.
- **driver.get(...) —** Opens the SauceDemo login page.
- **By.ID —** Finds webpage elements using their HTML id values.
- **send_keys() —** Types the username and password into the input fields.
- **click() —** Clicks the Login button.
- **assert —** Checks that the current URL contains inventory.html, indicating that login reached the inventory page.
- **try / except / finally —** Handles errors, prints PASS or FAIL, and closes the browser at the end.

6. Screenshots

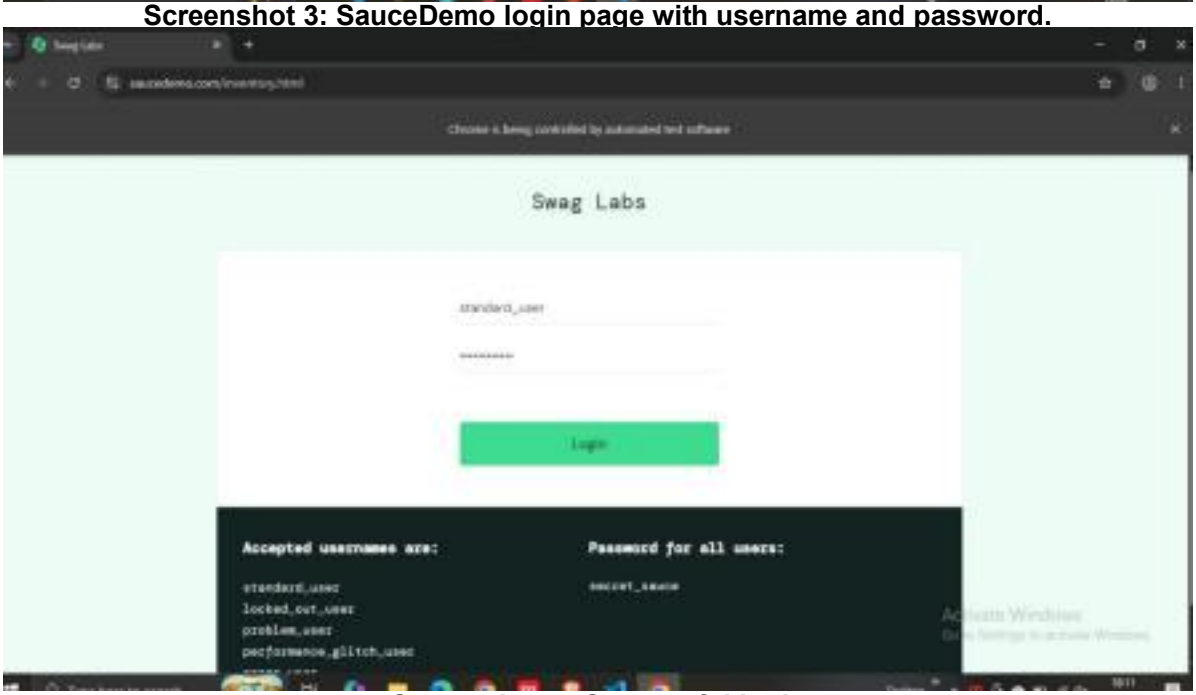
Screenshot 1: Python code in VS Code (first part).



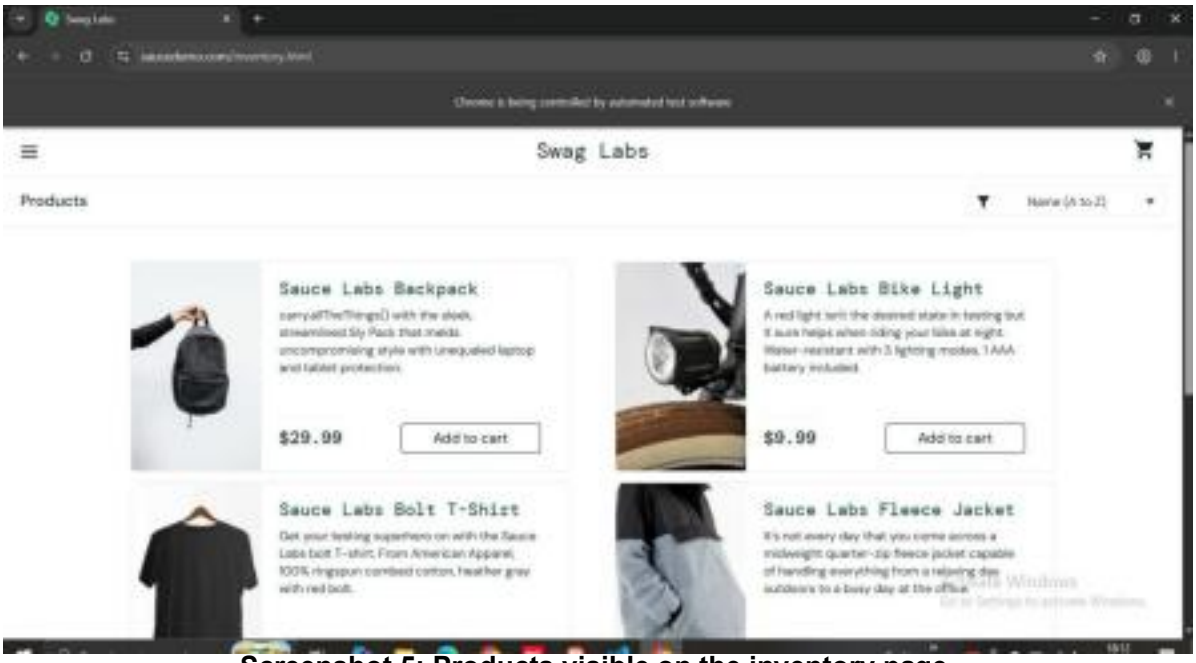
Screenshot 2: Python code in VS Code (validation and closing section).



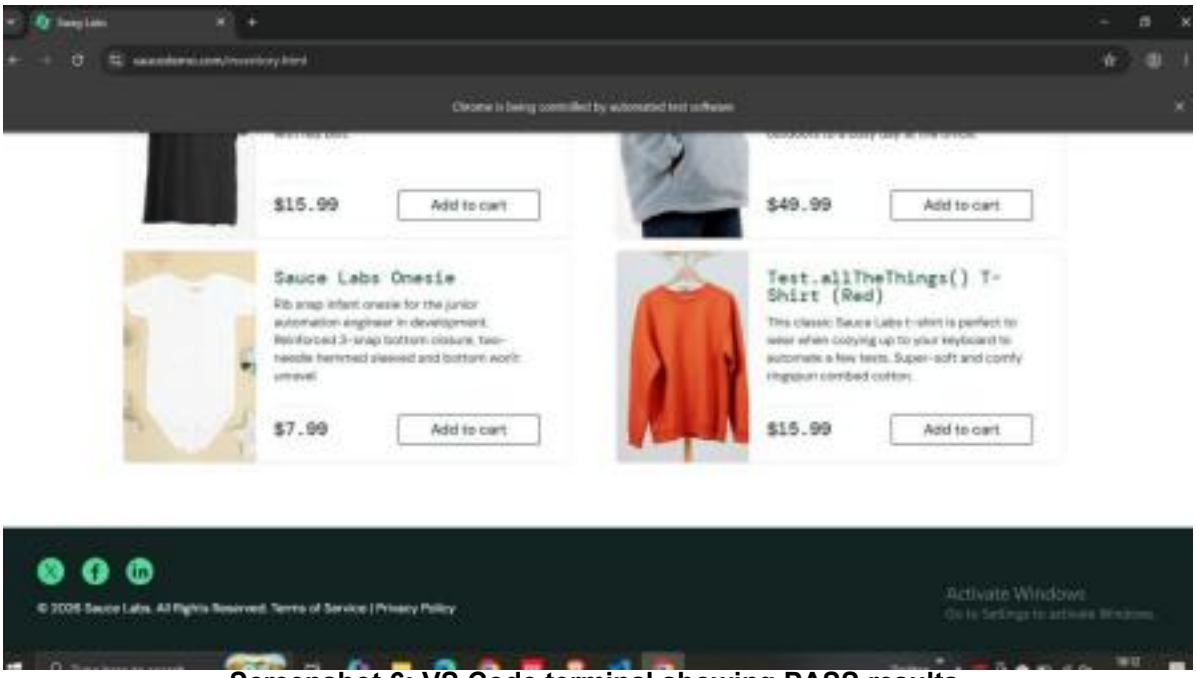
Screenshot 3: SauceDemo login page with username and password.



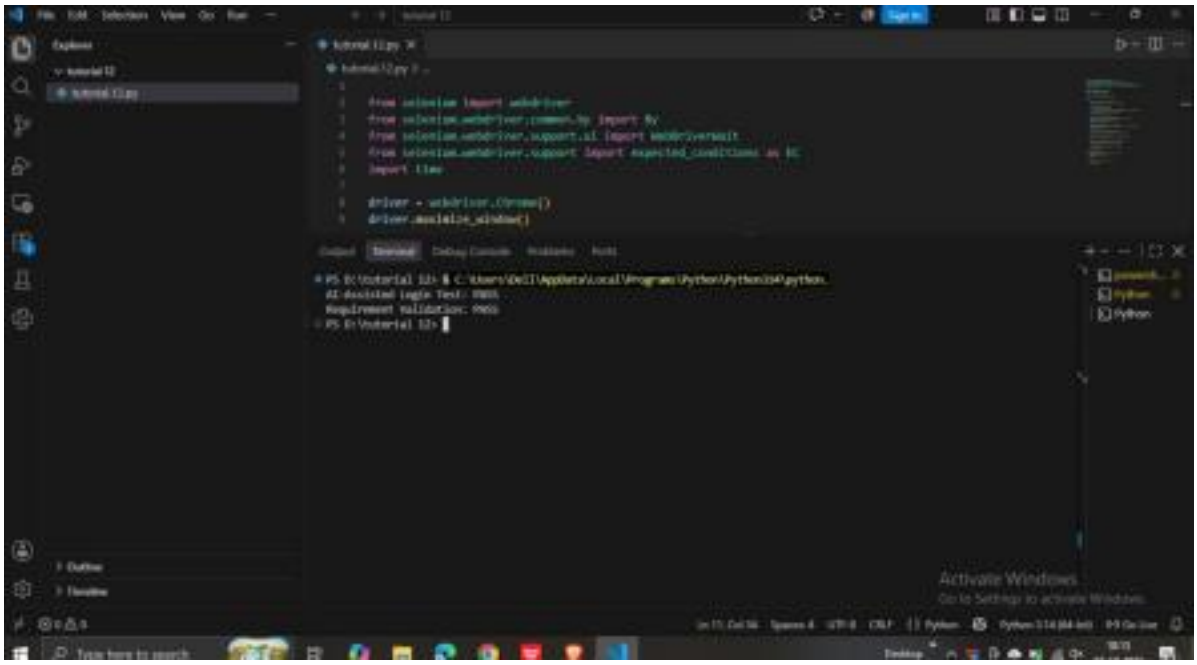
Screenshot 4: Successful login



Screenshot 5: Products visible on the inventory page.



Screenshot 6: VS Code terminal showing PASS results.



7. Expected output

```
AI-Assisted Login Test: PASS
Requirement Validation: PASS
```

8. Result / Conclusion

The login test was successfully automated using Python and Selenium. The program entered the test credentials, clicked Login, verified that the inventory page opened, displayed PASS messages, and closed the browser.